MARIAM OUDEH

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SUMMARY

Computer Engineering senior with hands-on experience building and debugging embedded electronics, including designing circuit boards, writing firmware, and fixing communication issues between hardware components. Comfortable setting up lab equipment and working independently on embedded hardware projects from start to finish. Strong background in C, C++, and Python, with experience turning technical work into clear documentation and reports.

EDUCATION

B.S. Computer Engineering | California State University, Sacramento *Expected Dec 2026*

A.S. Computer Science | American River College *2024*

RELEVANT PROJECTS

Senior Capstone: Automated Hydroponic Greenhouse (STM32H753ZIT6) *Spring–Fall 2026*

- Designed the circuit board layout for the project's control, motor, power, boards using KiCad 10, connecting the microcontroller to all the sensors, displays, and other parts it needs to talk to.
- Lead the lighting system for the project: wrote the firmware that controls brightness across 7 LED channels using PWM and MOSFETs, with different light settings for different plants.
- Found and fixed communication problems between boards using UART and SPI, the two methods the boards use to send data back and forth to each other.
- Designed and ordered a circuit board for the lighting system in EasyEDA, then had it manufactured through JLCPCB.

Wearable Glucose Monitoring Bracelet (Personal Project) *Feb 2026–Present*

- Building a wearable on an ESP32-S3 that listens in on BLE signals broadcast by a Dexcom G7 glucose sensor, using the ESP32's built-in Bluetooth radio to pick up the readings.

Bluetooth-Controlled RC Car (STM32) *2025*

- Physically built the car's circuit, wiring together a 4xAA battery pack, a motor driver, a Bluetooth UART Friend module, and an STM32 board.
- Wrote firmware that lets the Bluetooth module send commands to the STM32 over UART, which the STM32 then turns into signals that drive the motors.

Skystar E-Commerce Site *2026*

- Built and deployed a full e-commerce website for a small business, including product listings, site design, and front-end functionality.
- Designed a custom admin interface so the non-technical business owner can independently add and update products and events.

Personal Portfolio Website *2026*

- Built and designed a personal portfolio site from scratch in HTML/CSS/JavaScript to showcase embedded systems and biomedical hardware projects to employers.

EXPERIENCE

Outreach Specialist / Mentor | TRIO ETS, American River College, Sacramento, CA *Nov 2022–Present*

- Maintain and audit a student database of roughly 1,500 records, generating federal compliance reports on tight deadlines.
- Coordinate field trips and STEM workshops for first-generation middle and high school students, managing logistics, permissions, and vendor communication end to end.
- Produce technical documentation, presentations, and reports for internal and federal review.

SKILLS

Languages/Firmware: C, C++, Python, Java, Verilog, Embedded C/HAL

Protocols/Interfaces: UART, SPI, BLE, PWM, USB CDC

Hardware/Tools: STM32 (STM32H7, STM32L4), ESP32-S3, KiCad, EasyEDA, oscilloscope, logic analyzer, schematic capture, PCB layout, lab equipment/OS setup (Windows/Linux)

Other: Technical documentation, data analysis, cross-team communication

ACTIVITIES

Girls Who Code | Society of Women Engineers | Forensic Engineering Club | Muslim Student Association